A candidate exact two-soliton S-matrix for the bad Boussinesq equation

Project *quantum_boussinesq_solitons* (Phase 2, next step 3)

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Abstract

We construct a candidate exact quantum two-soliton S-matrix for $v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx}$ and test it against everything Phase 2 established. The candidate is a ratio of Gamma functions in the *spectral-parameter difference* $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$, times a power of a cross-ratio of the cubic's roots, times the exact topological factor:

$$S_{\rm ov} = \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)}\Big(\frac{e^2 A_{12}}{\eta^2}\Big)^{i\eta} e^{\frac{i}{\hbar}\frac{2}{3}M_1M_2(v_1-v_2)}, \qquad S_{\rm ho} = \frac{\Gamma(1-i\eta)}{\Gamma(1+i\eta)}\Big(\frac{A_{12}\eta^2}{e^2}\Big)^{i\eta} e^{\frac{i}{\hbar}\frac{2}{3}M_1M_2(v_1-v_2)}.$$

It reproduces the semiclassical phase δ (including the topological term) up to a constant Maslov-type phase $\pm\pi/2$; it is exactly unitary, real-analytic, and obeys the crossing relation exactly (this fixes $S_{\rm ho}$). On the physical sheet its only poles at breather kinematics are simple poles exactly at the Bogdan–Charkina Bohr–Sommerfeld points $I_{\rm br} = N\hbar$, $N \geq 1$. Fused breather–soliton amplitudes are unitary with the right classical limit. Three other families are *excluded* at leading order: Gamma functions of root differences, any product $F(\beta_1-\beta_2)G(\beta_1+\beta_2)$ (so the pure $\mathfrak{sl}_3$ "mirror" form, and quantum dilogarithms of $\beta_1 \pm \beta_2$), and the pure exponential $e^{i\delta/\hbar}$ (no bound states). The candidate is a guess **(d)** whose properties are derived **(c)**; it is *not* derived from a quantum theory. Its remaining ambiguities are all $O(\hbar^0)$ in the phase and are listed with the input that would fix them.

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1 Setting and requirements

Conventions are those of `report.pdf`: $k = \sinh\beta$, $v = \epsilon\cosh\beta$, $\ell = \epsilon\cosh 3\beta$, $M = 2k$, $P = \frac{2}{3}\epsilon k^3\cosh\beta$; particle 1 starts on the left. The semiclassical phase is **(c)**

$$\delta(1,2) = \tfrac{2}{9}(\ell_1 - \ell_2)\log A_{12} + T_{12}, \qquad T_{12} = \tfrac{2}{3}M_1M_2(v_1 - v_2), \tag{1}$$

with $S \simeq e^{i\delta/\hbar}$. An exact candidate must (i) reproduce (1) as $\hbar \to 0$, (ii) be unitary, (iii) obey the crossing-type relation $S_{\rm ov}(\beta_1, \beta_2 + i\pi)S_{\rm ho}(\beta_1, \beta_2) = 1$ (the exact version of $\delta_{\rm ov}(\beta_1, \beta_2 + i\pi) = -\delta_{\rm ho}(\beta_1, \beta_2)$), (iv) have bound-state poles consistent with the string condition $\ell_1 - \ell_2 \in \mp\frac{9i}{2}\hbar\mathbb{N}$, equivalently with the Bogdan–Charkina (BC) spectrum $I_{\rm br} = N\hbar$ **(a)** [1], and (v) give consistent fused amplitudes. Scattering is diagonal, so Yang–Baxter is automatic.

Roots of the cubic. **(c)** (`scripts/smatrix_ansatz.py`) The three roots attached to a right-mover are

$$\mu_j = \tfrac{i}{\sqrt{3}}\cosh\!\big(\beta + \tfrac{i\pi(2j+1)}{3}\big), \quad j = -1, 0, 1,$$

roots of $\mu^3 + \frac{1}{4}\mu = \frac{i}{12\sqrt{3}}\ell$; the first two are the pair $\frac{1}{2}(iv/\sqrt{3} \pm k)$ of Phase 2, whose cross-ratio is A_{12}. Consequently, for every root μ_{1a} of soliton 1,

$$\prod_b (\mu_{1a} - \mu_{2b}) = \tfrac{i}{12\sqrt{3}}(\ell_1 - \ell_2). \tag{2}$$

So $\ell_1 - \ell_2$ is itself a product of root differences, and every function below is a product of powers of root differences.

2 Families tested and excluded

Single-particle factors are a convention, so we compare the mixed derivative $\partial_{\beta_1}\partial_{\beta_2}$ of the leading phase with that of δ, by least squares at 40 random overtaking points (`scripts/smatrix_rejected_forms.py`). **(c)**

Family	Leading phase spans	Residual
$\prod\Gamma(1+ic\,x_{ab}/\hbar)^{n_{ab}}$, $x_{ab} = \mu_{1a} - \mu_{2b}$ (any complex c, n), $\times e^{iT/\hbar}$	$\Re, \Im\; x\log x$, x, T	2×10^{-2}: excluded
$F(\beta_1 - \beta_2)\,G(\beta_1 + \beta_2)$: $\mathfrak{sl}_3$ Toda / sine-Gordon mirror blocks, quantum dilogarithms of $\beta_1 \pm \beta_2$	$f(D) + g(S)$, so $\partial_S\partial_D = 0$	$\partial_S\partial_D\delta \neq 0$, e.g. -0.55 at $(S, D) = (1.4, 0.6)$: excluded
$e^{i\delta/\hbar}$ exactly	δ	passes (i)–(iii), *no poles*: excluded by (iv)
Control: Γ in η (Sec. 3)	$(\ell_1 - \ell_2)\log x_{ab}$, T	5×10^{-16}

The obstruction to the first two is the same: δ has the *uniform* weight $\ell_1 - \ell_2$ in front of every logarithm, while Stirling asymptotics of $\Gamma(\ldots + iX/\hbar)$ give weight X in front of $\log X$. The weight must therefore be the Gamma argument. This forces the Gamma functions to depend on $\eta \propto \ell_1 - \ell_2$ only. **(c)**The quantum-dilogarithm case is excluded only for arguments that are functions of $\beta_1 \pm \beta_2$ separately; a dilogarithm with mixed arguments was not tested.

3 The candidate

Let $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and $\mathcal{A}_{12} = \dfrac{81 A_{12}}{4(\ell_1 - \ell_2)^2}$, so $A_{12}/\eta^2 = \hbar^2 \mathcal{A}_{12}$. **(d)**

$$S_{\epsilon\epsilon}(1,2) = \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} \left(e^2\hbar^2 \mathcal{A}_{12}\right)^{i\eta} e^{iT_{12}/\hbar} \qquad (\epsilon_1 = \epsilon_2), \tag{3}$$

$$S_{+-}(1,2) = \frac{\Gamma(1-i\eta)}{\Gamma(1+i\eta)} \left(\frac{A_{12}\eta^2}{e^2}\right)^{i\eta} e^{iT_{12}/\hbar} \qquad \text{(head-on)}. \tag{4}$$

Proposition 3.1 (**(c)**, `scripts/smatrix_ansatz.py`)**.** *1. $\mathcal{A}_{12}$ is finite and nonzero at threshold $\beta_1 = \beta_2$, so (3) has no singularity there: the $\log(\ell_1 - \ell_2)^2$ singularity of $\log A_{12}$ is carried by the Gamma ratio. Using (2), $\mathcal{A}_{12}$ is a ratio of products of root differences.*

2. $\log \mathcal{A}_{12} = h(\beta_1 + \beta_2) + g(\beta_1 - \beta_2)$: the $\mathfrak{sl}_3$/sine-Gordon blocks of Phase 2 (Prop. 5.1 of `report.pdf`) survive, but only in the exponent*, multiplied by η.*

3. $\Gamma(ix)/\Gamma(-ix) = -\Gamma(1+ix)/\Gamma(1-ix)$. Hence the Gamma part is that of the quantum Toda two-body amplitude $R = \Gamma(i\kappa)/\Gamma(-i\kappa)\hbar^{2i\kappa}$ (`toda_route.pdf`), with $\kappa = \Delta p/\hbar \mapsto \eta$, and the Toda Lax eigenvalue $p \mapsto \frac{2}{9}\ell$. That p maps to ℓ in the phase weight was found in Phase 2b; that the Gamma structure survives the Boussinesq limit is **(d)**.

4 Tests

4.1 Semiclassical limit, unitarity, crossing

Proposition 4.1 (**(c)**, `scripts/smatrix_semiclassical_unitarity_crossing.py`)**.** *For real rapidities, at three overtaking and three head-on points:*

1. $\arg S - \delta/\hbar \to \frac{\pi}{2}\sigma \operatorname{sgn}\eta$ (mod 2π), $\sigma = +1$ (same direction), -1 (head-on), with remainder $O(\hbar)$; at $\hbar = 10^{-2}, 10^{-3}, 10^{-4}$ the remainder is e.g. $-6.4\times10^{-4}, -6.4\times10^{-5}, -6.4\times10^{-6}$. Without the factor $e^{iT/\hbar}$ the remainder is $O(1/\hbar)$, so the topological term is required.

2. $|S| = 1$ and $S(1,2)S(2,1) = 1$ to 30 digits.

3. $S_{\rm ov}(\beta_1, \beta_2 + i\pi)\, S_{\rm ho}(\beta_1, \beta_2) = 1$ to 10^{-27}; at leading order this is $\delta_{\rm ov}(\beta_1, \beta_2 + i\pi) + \delta_{\rm ho} = 0$.

4. $S(\bar\beta_1, \bar\beta_2) = 1/\overline{S(\beta_1, \beta_2)}$ at generic complex rapidities.

Crossing is what fixes (4): under $\beta_2 \to \beta_2 + i\pi$, $\eta_{\rm ov} \to \eta_{\rm ho}$ and $A_{\rm ov} \to 1/A_{\rm ho}$ (Phase 2), so the inverse Gamma ratio and the power $(A\eta^2/e^2)^{i\eta}$ are forced. Using (3) for head-on scattering too would violate crossing. During the work we first wrote (4) with $(\eta^2/e^2A)^{i\eta}$; the semiclassical test failed, and crossing gave the correct form.

The constant $\pm\pi/2$ is a Maslov-type phase at $O(\hbar^0)$. The report listed a missing Maslov phase among its least-sure points; the candidate predicts one. **(d)**

4.2 Poles and the breather spectrum

Take two right-movers at $\beta_{1,2} = a \mp ib$ with $\pi/3 < b < \pi/2$ (these breathers are supersonic and move to the *left*, with envelope velocity $V = \cosh a \cos 2b/\cos b < -\cosh a$; the range $\pi/2 < b < 2\pi/3$ gives their right-moving parity images. An earlier version of this note called them right-moving; corrected in `oneloop.pdf`).

Proposition 4.2 (**(c)**, `scripts/smatrix_poles_bootstrap.py`)**.** *1. With this labelling $\operatorname{Im}(P_1 - P_2) > 0$: the relative momentum lies in the upper half plane, so this is the physical sheet for bound states of $S(1,2)$. Total E and P are real.*

2. *$i\eta = \frac{4}{9}\sinh 3a \sin 3b/\hbar = -I_{\rm br}/\hbar$, with $I_{\rm br} = -\frac{4}{9}\sinh 3a \sin 3b$ the BC adiabatic invariant. Hence $S(1,2)$ has simple poles (from $\Gamma(1+i\eta)$) exactly at $I_{\rm br} = N\hbar$, $N = 1, 2, \ldots$: the BC Bohr–Sommerfeld spectrum, with their N (not $N + \frac{1}{2}$). Verified for $N \leq 4$ at $a = 0.5, 0.8$, $\hbar = 0.1$ ($|S||\delta b|$ constant to 10^{-9}).*

3. *There is no pole at $N = 0$ (threshold $b = \pi/3$); there the Toda form $\Gamma(i\eta)$ would have one. This selects $\Gamma(1 + i\eta)$.*

4. *The form that is regular at real threshold and the form with poles on the physical sheet coincide. This was not imposed and is a non-trivial consistency check.*

5. *At a pole $e^{i\eta \cdot 2\pi i} = 1$, so the residue modulus does not depend on the branch of $\log \mathcal{A}_{12}$ (which lies on its cut, $\mathcal{A}_{12} < 0$, at breather kinematics). The residue phase does.*

Sample energies $E_N = E(\beta_1) + E(\beta_2)$ at $\hbar = 0.1$, $a = 0.8$: $0.8711, 0.9759, 1.0759, 1.1705$ for $N = 1..4$ (**(c)**, numerical; identical to BC's quantisation because the conditions coincide).

4.3 Bootstrap: breather–soliton amplitudes

Proposition 4.3 (**(c)**, `scripts/smatrix_poles_bootstrap.py`)**.** *Define $S_{B_N,3} = S(1,3)S(2,3)$ at the N-th pole. Then:*

1. *$|S_{B_N,3}| = 1$ for real β_3 of either direction, and $\delta_{13} + \delta_{23}$ is real (the classical breather–soliton phase, by Hirota additivity).*

2. *$\arg S_{B_N,3} - (\delta_{13} + \delta_{23})/\hbar \to \pi$ as $\hbar = I_{\rm br}/N \to 0$ at fixed (a, b): two Maslov-type constants.*

3. *With $x_c = \frac{i}{2}(\eta_{13} + \eta_{23})$, the Gamma part telescopes to*

$$\frac{\Gamma(1 + x_c - \frac{N}{2})\Gamma(1 + x_c + \frac{N}{2})}{\Gamma(1 - x_c + \frac{N}{2})\Gamma(1 - x_c - \frac{N}{2})}.$$

It has a simple pole at $x_c = -\frac{N}{2}$, double poles at $x_c = -\frac{N}{2} - 1 - m$, and the mirror-image zeros; $N - 1$ pole–zero pairs cancel in between. This was checked numerically for $N = 1, 2, 3$.

Interpretation **(d)**: the simple pole is at $\ell_3 = \ell_2$, $\ell_1 - \ell_3 = N$-string, an exchange of soliton 3 with a constituent. The double poles extend the string, $\ell_3 = \ell_2 - \frac{9i}{2}(m + 1)\hbar$, and would need a Coleman–Thun-type explanation. Whether these lie on the physical sheet of the three-body kinematics was *not* determined, so the bootstrap is only partially tested. Breather–breather amplitudes follow by the same fusion but were not analysed.

5 Summary of tests

Requirement	Result for (3)–(4)	Status
Semiclassical limit	$\delta/\hbar \pm \pi/2 + O(\hbar)$, topological term needed	**(c)** passed
Unitarity	exact	**(c)** passed
Crossing	exact; fixes the head-on form	**(c)** passed
Real analyticity	exact	**(c)** passed
Poles vs string / BC	simple poles exactly at $I_{\rm br} = N\hbar$, physical sheet, none at $N = 0$	**(c)** passed
Bootstrap	fused amplitudes unitary, right classical limit; pole interpretation open	**(c)** partial
Derivation from a quantum theory	none	**(d)**

6 Remaining ambiguities and what would fix them

All tests constrain the phase only at orders $\hbar^{-1}$ (fixed by δ) and at the pole positions. The following modifications pass every test above, or fail only the poles in a way the present data cannot decide. **(c)** (that they pass) / **(d)** (that this list is complete)

1. **CDD factors.** Any $f(1,2)$ with $f(1,2)f(2,1)=1$, $|f|=1$ for real rapidities, $f_{\rm ov}(\beta_1,\beta_2+i\pi)f_{\rm ho}(\beta_1,\beta_2)=1$, and $\log f = O(\hbar^0)$. Examples: $\prod_j \frac{\sinh\frac{\pi}{2}(\eta-i\alpha_j)}{\sinh\frac{\pi}{2}(\eta+i\alpha_j)}$, or $\frac{c+i\eta}{c-i\eta}$ with constant c. They change one-loop phases and add or remove poles; the latter could be tested against a quantum breather spectrum.

2. **Renormalisation of** $\hbar$ in η, $\hbar \to \hbar_{\rm ren} = \hbar(1+c_1\hbar+\ldots)$, as in sine-Gordon $\beta^2 \to \beta^2/(1-\beta^2/8\pi)$. This shifts the breather spectrum at $O(\hbar^2)$.

3. **Shift of the string**: $\Gamma(\frac{1}{2}+i\eta)$ instead of $\Gamma(1+i\eta)$, i.e. $I_{\rm br}=(N-\frac{1}{2})\hbar$. This is excluded only if one trusts BC's integer quantisation, which is itself a leading-order Bohr–Sommerfeld statement.

4. $O(\hbar)$ **corrections to** $\mathcal{A}_{12}$ inside the power, and the branch of $\log\mathcal{A}_{12}$ at complex rapidities (residue phases).

5. **The Maslov-type constant** $\pm\frac{\pi}{2}$, and whether the overall sign of S is a convention.

Input that would fix them.

- *One-loop soliton and breather spectrum* (DHN-type; next step 2 of the report). The $O(\hbar^0)$ correction to the breather energies E_N fixes items 2–3. The one-loop two-soliton phase (from the phonon phase shifts on the reflectionless soliton background) fixes item 4 and tests the $\pm\pi/2$.

- *Small-amplitude Toda limit* (next step 1: the Fermi-edge double scaling $\beta\to 0$, $\hbar\propto\beta^2$, of the quantum Toda Bethe ansatz). This gives an exact quantum S-matrix in the corner $\beta\to 0$ against which (3)–(4), including possible CDD factors, can be compared directly. Note $\eta=O(1)$ there means $\hbar\propto\beta^4$ for overtaking ($\ell_1-\ell_2\approx\frac{9}{2}(\beta_1^2-\beta_2^2)$) but $\hbar=O(1)$ for head-on ($\ell_1-\ell_2\approx 2$). So the head-on Gamma ratio is never semiclassical in that corner. **(c)** (elementary)

- *A quantum lattice regularisation* at the dispersion scale would be needed to derive the candidate rather than guess it; none is known (`toda_route.pdf`).

A Scripts

All are in `phase2/scripts`; each ends with `ALL CHECKS PASSED`.

- `smatrix_ansatz.py`: definitions; roots, Eq. (2), Prop. 3.1.
- `smatrix_semiclassical_unitarity_crossing.py`: Prop. 4.1.
- `smatrix_poles_bootstrap.py`: Props 4.2–4.3.
- `smatrix_rejected_forms.py`: Section 2.

Failures along the way:

- The first root labelling mismatched the Phase 2 pair. It was fixed by matching to `consistency_checks.py`.

- The first head-on form had the wrong sign of the A-power; it failed the semiclassical and crossing tests and was corrected as described above.
- A threshold-size tolerance was set too tightly ($|\mathcal{A}| > 10^{-3}$, whereas $\mathcal{A} \approx 2 \times 10^{-4}$ at $\beta = 1.6$).
- $|S|$ at complex breather kinematics is very large ($\sim 10^{23}$ at $a = 0.8$) because $|A^{i\eta}| \neq 1$ there. The pole test therefore uses scaling with the distance to the pole, not absolute size.

References

[1] M. M. Bogdan, O. V. Charkina, arXiv:2502.14695, *Low Temp. Phys.* 51 (2025).